

HOWARD UNIVERSITY · DEPARTMENT OF MATHEMATICS

Homework: Outliers

MATH 153 · Data Wrangling and Data Engineering

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Due: Saturday, October 3, 2026, on Canvas

Two questions. Question 1 needs no computation, only short written answers. Question 2 is a small hand calculation on fifteen numbers; you may use Python for the arithmetic, but show your numbers either way.

Question 1. Name the kind, then say what you would do

A hospital clinic keeps a table of patient visits. Here are five things you find in it. For each one, write down (i) which label it is (*error*, *contamination*, *genuine extreme*, or *inconsistency*) and (ii) in a few words, what you would do about it.

- (a) `blood_pressure = -1` in a file where the intake form writes `-1` when the reading was not taken.
- (b) A wait of 100 minutes. The front desk confirms the patient really did wait that long: the only doctor on duty was called into surgery.
- (c) One row has `visit_date = 03/04/2026`. Every other row in that column looks like `2026-03-04`.
- (d) A 3-day-old infant appears in a table of adult cardiology patients.
- (e) The `department` column contains `Cardiology`, `cardiology`, and `" Cardiology"` (note the leading space), all for the same department.

Then answer these two:

- (f) Exactly two of the five are inconsistencies. Which two? Then explain why a *z*-score run on a numeric column could never find either of them.
- (g) The lecture said: *Removing a point needs a cause, not just a distance*. Explain in one sentence what goes wrong if you ignore that, using item (b) as your example.

Question 2. Run all three detection rules on the same fifteen numbers

Here are the waiting times, in minutes, for fifteen patients on one morning:

14, 15, 13, 14, 16, 13, 15, 14, 14, 15, 13, 14, 16, 14, 100

Sort them first; you will need the sorted list for parts (b) and (c).

- (a) **The z -score.** Compute the mean $\bar{x}$ and the sample standard deviation s (`ddof=1`, divide by $n - 1$). Then compute

$$z = \frac{100 - \bar{x}}{s}.$$

Does the rule $|z| > 3$ flag the 100? Also compute the ceiling $\frac{n-1}{\sqrt{n}}$ for $n = 15$, and say what fraction of it your z reached.

- (b) **The IQR rule.** With 15 sorted values, Q_1 is the 4th value and Q_3 is the 11th, so no interpolation is needed.

- (i) Write down Q_1 , Q_3 , and $\text{IQR} = Q_3 - Q_1$.
- (ii) Compute the two fences $Q_1 - 1.5 \text{ IQR}$ and $Q_3 + 1.5 \text{ IQR}$.
- (iii) Which values fall outside them?

- (c) **The modified z -score.** Recall

$$M_i = \frac{0.6745(x_i - \tilde{x})}{\text{MAD}}, \quad \text{flag when } |M_i| > 3.5,$$

where $\tilde{x}$ is the median and $\text{MAD} = \text{median } |x_i - \tilde{x}|$.

- (i) Write down the median $\tilde{x}$ (the 8th of the 15 sorted values).
 - (ii) List the absolute deviations $|x_i - \tilde{x}|$, sort that list, and read off its median. That is the MAD.
 - (iii) Compute M for the value 100. Is it flagged?
- (d) **One sentence.** All three rules agree here. Using your four labels from Question 1, can any of the three rules tell you *which kind* of outlier the 100 is? Say why or why not.

Useful NumPy, if you use Python for Question 2: `x.mean()`, `x.std(ddof=1)`, `np.median(x)`. For the MAD: `m = np.median(x)`, then `np.median(np.abs(x - m))`.

Remember: the MAD is the *median* of the absolute deviations, not their mean.

Submission

- One PDF on Canvas, questions answered in order and clearly labeled.
- If you used Python, include your code and its output.
- Handwritten work is accepted if it is scanned clearly.
- Discuss ideas with classmates, but write up your own answers.